



Keep MOR Thin

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Morpheus does not need to become large to become real. The heavy machinery already runs. MOR taps it. It does not rebuild it.

The surface, in their numbers. Four subsystems decide whether Morpheus is real: inference, agents, verification, settlement. A sovereign L2 builds and secures all four. Five people ship every native line of Morpheus [verified], MOR trades at \$1.95 on a \$17M cap with \$8.9k of daily volume and one live use, farming emissions, and settlement is `pricePerSecond` times elapsed on a provider-signed receipt with no slashing [verified]. The deeper limit is architectural: `api.mor.org` funds every Web2 caller from one shared consumer wallet, and after a nonce conflict opens and closes serialize on a wallet lock for 20 seconds (`config.py:319-323`) [verified]. One wallet, one nonce, a few sessions a second, beneath a provider ceiling of a few hundred concurrent sessions [verified]. Rebuilding all four to fix a settlement rule is the expensive way to buy a receipt.

Keep MOR thin. MOR stays a budget and a reward. The heavy functionality ships on native chains Morpheus taps, not code it writes.

- **Inference and proof.** The A-Chain (`lux/ai`, `lux/chains/aivm`): bonded operators, commit-reveal quorum, `SettleQuorum`, `SlashProvider`, local nvtrust/SPDM/RIM attestation, no cloud [verified]. The quorum produces the verified-work receipt; MOR settles against it.
- **Agents.** The Hanzo agentVM: trajectories, tool loops, memory, and budget-capped spawning (LP-325 linear budgets, LP-322 coordination primitive) [verified].
- **Settlement and finality.** Lux primary, post-quantum through Quasar (Horizon) [verified]. **Governance.** Zoo, plus DeSci [verified].

An agent runs on the agentVM, verifies inference on the A-Chain, settles its budget in MOR. Morpheus writes little new code. It interweaves.

The outcome, projected. Two results, kept separate. The core stays small and auditable: token, treasury, rewards, and the thin settlement it already ships. And MOR gains a use that scales with verified work, not deposits. `mord` measures the ceiling: settlement arithmetic costs about $2\mu\text{s}$ per session, while the un-fakeable cost, two ed25519 verifies per meter, is about $296\mu\text{s}$, so roughly 3,400 verified sessions/sec/core and 25k/sec at eight cores [verified, measured 2026-08-29, M1 Max, single core, in-memory; `mord/BENCHMARKS.md`]. Honest boundary: that is the settlement-plus-verification rate, not end-to-end network TPS, which consensus and networking set [verified]. But the single-wallet wall is gone, faked work earns zero, model substitution is slashed, and 40 tests pass [verified]. Best case, assumptions stated: if agent budgets settle in MOR against A-Chain receipts, MOR demand tracks verified sessions across four live networks rather than the emission schedule [illustrative], feeding the \$511K buyback Morpheus already runs with real volume rather than yield [verified].

Contrast, honestly (ADOPT-01). The full L2 buys more and costs more: a sovereign validator set, MOR as native gas, and fee policy under Morpheus governance, against building and securing all four subsystems and funding validators outside the Lux staking economy. The trade is sovereignty for surface area.

Relevance, and first step. The substantial engineering is done and running on our side: A-Chain quorum and slashing, the agentVM, Quasar finality, the Enso router and `/v1` surface, metering and `olly` [verified]; `12-template` stands an interweave point up in minutes [verified]. No other counterparty can lend a verified-inference chain, an agent runtime, post-quantum settlement, and a governance network that all run today. First step, and it commits nothing: point one agent workload at the agentVM, verify one session on the A-Chain, settle one budget in MOR. The rest composes onto that.

ADOPT-12, the thin alternative to ADOPT-01. A-Chain in `lux/ai` and `lux/chains/aivm`; agentVM in LP-322/LP-325; Quasar in `luxfi/quasar`; benchmark in `lux/mord/BENCHMARKS.md`. Citations checked against the working tree. • `research@lux.network`